Mechanics of Flexible Structures & Soft Robots
MAE-263F
Fall 2026

MAE-263F Sample Midterm 1: Solutions

This sample midterm has 29 problems. The actual midterm will have 20 problems, each worth 0.5 pts.

Question 1

Newton's method is used to solve $f(x) = x^2 - 3x + 2 = 0$ with the initial guess $x_0 = 0$. What is x_1 after **one** Newton iteration?

(a) $x_1 = 1$

(b) $\boldsymbol{x_1 = 2/3}$ ✓

(c) $x_1 = -2/3$

(d) $x_1 = 3/2$

Answer: (b)
Solution. $f(0) = 2$ and $f'(x) = 2x - 3$, so $f'(0) = -3$. Then $x_1 = x_0 - f(x_0)/f'(x_0) = 0 - 2/(-3) = 2/3$.

Question 2

Consider the system of equations

$$f_1 \equiv x_1^3 + x_2 = 0, \qquad f_2 \equiv -x_1 + x_2^3 + 1 = 0.$$

What is the Jacobian $\mathbf{J}$, with $J_{ij} = \partial f_i / \partial x_j$, evaluated at $(x_1, x_2) = (1, 2)$?

(a) $\begin{bmatrix} 3 & -1 \\ 1 & 12 \end{bmatrix}$

(b) $\begin{bmatrix} 3 & 1 \\ -1 & 6 \end{bmatrix}$

(c) $\begin{bmatrix} \mathbf{3} & \mathbf{1} \\ \mathbf{-1} & \mathbf{12} \end{bmatrix}$ ✓

(d) $\begin{bmatrix} 1 & 1 \\ -1 & 8 \end{bmatrix}$

Answer: (c)
Solution. $\mathbf{J} = \begin{bmatrix} \partial f_1/\partial x_1 & \partial f_1/\partial x_2 \\ \partial f_2/\partial x_1 & \partial f_2/\partial x_2 \end{bmatrix} = \begin{bmatrix} 3x_1^2 & 1 \\ -1 & 3x_2^2 \end{bmatrix}$. At $(1, 2)$: $\mathbf{J} = \begin{bmatrix} 3 & 1 \\ -1 & 12 \end{bmatrix}$. (Option (a) is the transpose.)

Question 3

Consider the system of equations

$$f_1 \equiv x_1^2 - x_2 = 0, \qquad f_2 \equiv x_1 + x_2 - 2 = 0.$$

Starting from the initial guess $\mathbf{x}_0 = [0,\ 0]^T$, what is $\mathbf{x}_1$ after **one** Newton iteration, $\mathbf{x}_1 = \mathbf{x}_0 - \mathbf{J}^{-1}\mathbf{f}$?

(a) $\mathbf{x_1 = [2,\ 0]^T}$ ✓

(b) $\mathbf{x}_1 = [-2,\ 0]^T$

(c) $\mathbf{x}_1 = [1,\ 1]^T$

(d) $\mathbf{x}_1 = [0,\ 2]^T$

Answer: (a)

Solution. At $\mathbf{x}_0 = [0,0]^T$: $\mathbf{f} = [0^2 - 0,\ 0 + 0 - 2]^T = [0, -2]^T$ and $\mathbf{J} = \begin{bmatrix} 2x_1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$. Solve $\mathbf{J}\Delta\mathbf{x} = \mathbf{f}$: row 1 gives $-\Delta x_2 = 0 \Rightarrow \Delta x_2 = 0$; row 2 gives $\Delta x_1 + \Delta x_2 = -2 \Rightarrow \Delta x_1 = -2$. Update: $\mathbf{x}_1 = \mathbf{x}_0 - \Delta\mathbf{x} = [0,0]^T - [-2,0]^T = [2,0]^T$. (Option (b) adds $\Delta\mathbf{x}$ instead of subtracting it.)

Question 4

For the same system as Question 3, $f_1 \equiv x_1^2 - x_2 = 0$ and $f_2 \equiv x_1 + x_2 - 2 = 0$, Newton's method fails at the first iteration if the Jacobian is singular ($\det \mathbf{J} = 0$). For which initial guess $\mathbf{x}_0 = [x_1,\ x_2]^T$ does this happen?

(a) $[0,\ 0]^T$

(b) $[1,\ 1]^T$

(c) $[0.5,\ -3]^T$

(d) $\mathbf{[-0.5,\ 3]^T}$ ✓

Answer: (d)

Solution. $\mathbf{J} = \begin{bmatrix} 2x_1 & -1 \\ 1 & 1 \end{bmatrix}$, so $\det \mathbf{J} = 2x_1 + 1$, which is zero when $x_1 = -0.5$ (for any x_2). For the other options, $\det \mathbf{J} = 1$, 3, and 2.

Question 5

Two masses, each with $m = 1$ kg, move along the x-axis and are connected by one linear spring with stiffness $k = 50$ N/m. Neither mass is fixed, and there is no damping or external force. The elastic energy is $E = \frac{1}{2}k(x_2 - x_1 - L)^2$, where L is the rest length. The system is simulated with implicit Euler and $\Delta t = 0.1$ s, so the Jacobian is $J_{ij} = \frac{m_i}{\Delta t^2}\delta_{ij} + \frac{\partial^2 E}{\partial x_i \partial x_j}$. What is $\det \mathbf{J}$?

(a) 0

(b) **20,000** ✓

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(c) 22,500

(d) 10,000

Answer: (b)

Solution. The Hessian of E is $k\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix} = \begin{bmatrix}50 & -50\\ -50 & 50\end{bmatrix}$ and $m/\Delta t^2 = 1/0.01 = 100$. Thus

$$\mathbf{J} = \begin{bmatrix}150 & -50\\ -50 & 150\end{bmatrix}, \qquad \det \mathbf{J} = 150^2 - 50^2 = 22{,}500 - 2{,}500 = 20{,}000.$$

The Hessian alone is singular ($\det = 0$), but the inertia term makes $\mathbf{J}$ invertible, so the dynamic simulation works even with no fixed DOFs.

Question 6

For the same two-mass system as Question 5, the Hessian of the elastic energy is

$$\mathbf{H} = \frac{\partial^2 E}{\partial \mathbf{x} \partial \mathbf{x}} = k\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}.$$

Which nonzero vector $\mathbf{v}$ lies in the null space of $\mathbf{H}$ (i.e., $\mathbf{Hv} = \mathbf{0}$)?

(a) $\mathbf{v = [1,\ 1]^T}$ ✓

(b) $\mathbf{v} = [1,\ -1]^T$

(c) $\mathbf{v} = [1,\ 0]^T$

(d) $\mathbf{v} = [0,\ 1]^T$

Answer: (a)

Solution. $\mathbf{H}[1,1]^T = k[1-1,\ -1+1]^T = \mathbf{0}$. Physically, moving both masses by the same amount (a rigid translation) does not change the spring length, so it does not change the elastic energy.

Question 7

What is the cross product $\mathbf{a} \times \mathbf{b}$, where $\mathbf{a} = [1,\ 1,\ 3]$ and $\mathbf{b} = [-2,\ 3,\ 5]$?

(a) $[4,\ 11,\ -5]$

(b) $[-4,\ 11,\ 5]$

(c) $\mathbf{[-4,\ -11,\ 5]}$ ✓

(d) $[14,\ -11,\ 5]$

Answer: (c)

Solution. $\mathbf{a}\times\mathbf{b} = [a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1] = [1\cdot 5 - 3\cdot 3,\ 3\cdot(-2) - 1\cdot 5,\ 1\cdot 3 - 1\cdot(-2)] = [-4, -11, 5]$. (Option (a) is $\mathbf{b}\times\mathbf{a}$.)

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Question 8

A 2D discrete beam has three nodes, $\mathbf{x}_1 = (0, 0)$, $\mathbf{x}_2 = (1, 0)$, and $\mathbf{x}_3 = (1, -1)$. Using the turning angle $\phi_2 = \tan^{-1} \frac{(\mathbf{e}^1 \times \mathbf{e}^2) \cdot \hat{\mathbf{e}}_z}{\mathbf{e}^1 \cdot \mathbf{e}^2}$ and the discrete curvature $\kappa_2 = 2 \tan(\phi_2/2)$, what is κ_2? (Evaluate $\tan^{-1}$ as the four-quadrant arctangent, atan2(numerator, denominator).)

(a) 2

(b) **−2** ✓

(c) 1

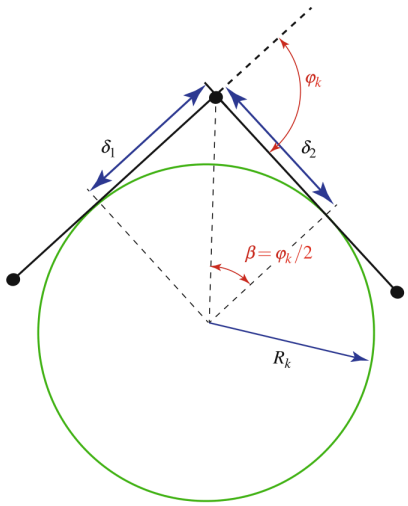
(d) $-\pi/2$

Answer: (b)
Solution. $\mathbf{e}^1 = (1, 0)$ and $\mathbf{e}^2 = (0, -1)$. The z-component of $\mathbf{e}^1 \times \mathbf{e}^2$ is $1 \cdot (-1) - 0 \cdot 0 = -1$, and $\mathbf{e}^1 \cdot \mathbf{e}^2 = 0$. Since the denominator is 0 and the numerator is negative, $\phi_2 = \text{atan2}(-1, 0) = -\pi/2$, and $\kappa_2 = 2 \tan(-\pi/4) = -2$. (Option (a) ignores the sign; option (d) is ϕ_2, not κ_2.)

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Question 9

The figure below shows two edges of a 2D discrete beam meeting at a node with turning angle φ_k. The circle (radius R_k) touches each edge at a distance $\delta_1 = \delta_2 = \delta$ from the node, and each radius drawn to a touching point is perpendicular to its edge.



If $\delta = 1$ m and $\varphi_k = 60°$, what is R_k?

(a) $1/\sqrt{3}$ m

(b) $2\sqrt{3}$ m

(c) 2 m

(d) **$\sqrt{3}$ m** ✓

Answer: (d)
Solution. In the right triangle formed by the center of the circle, a touching point, and the node, the angle at the center is $\beta = \varphi_k/2$, the side opposite to it is δ, and the side next to it is R_k. So $\tan(\varphi_k/2) = \delta/R_k$ and

$$R_k = \frac{\delta}{\tan(\varphi_k/2)} = \frac{1}{\tan 30°} = \frac{1}{1/\sqrt{3}} = \sqrt{3} \text{ m}.$$

When the circle touches each edge at its midpoint, $\delta = \Delta l/2$ and this becomes

$$R_k = \frac{\Delta l}{2\tan(\varphi_k/2)} = \frac{\Delta l}{\kappa_k}.$$

Question 10

A 2D discrete beam has three nodes, $\mathbf{x}_1 = (0,0)$, $\mathbf{x}_2 = (2,0)$, and $\mathbf{x}_3 = (2,2)$, with all lengths in meters. Both edges have the same length, $\Delta l = 2$ m. What is the radius R_2 of the osculating circle at $\mathbf{x}_2$, i.e., the circle that touches each edge at its midpoint?

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(a) **1 m** ✓

(b) 2 m

(c) 0.5 m

(d) 4 m

Answer: (a)
Solution. $\mathbf{e}^1 = (2, 0)$ and $\mathbf{e}^2 = (0, 2)$: the cross product (z) is 4 and the dot product is 0, so $\phi_2 = \pi/2$ and $\kappa_2 = 2\tan(\pi/4) = 2$. The circle touches each edge at its midpoint, so $R_2 = \Delta l/\kappa_2 = 2/2 = 1$ m (see Question 9 with $\delta = \Delta l/2 = 1$ m: $R_2 = 1/\tan 45° = 1$ m).

Question 11

A 2D discrete beam has three nodes, $\mathbf{x}_1 = (0, 0)$, $\mathbf{x}_2 = (1, 0)$, and $\mathbf{x}_3 = (1, 1)$, with all lengths in meters. The undeformed edge length is $\Delta l = 1$ m, the bending stiffness is $EI = 2$ N·m^2, and the natural curvature at $\mathbf{x}_2$ is $\kappa_2^0 = 1$. What is the bending energy $E_{b,2} = \frac{1}{2}\frac{EI}{\Delta l}(\kappa_2 - \kappa_2^0)^2$?

(a) 4 J

(b) 9 J

(c) **1 J** ✓

(d) 2 J

Answer: (c)
Solution. $\mathbf{e}^1 = (1, 0)$ and $\mathbf{e}^2 = (0, 1)$, so $\phi_2 = \pi/2$ and $\kappa_2 = 2$. $E_{b,2} = \frac{1}{2} \cdot \frac{2}{1} \cdot (2 - 1)^2 = 1$ J. (Option (a) ignores the natural curvature; option (b) uses $\kappa_2 + \kappa_2^0$.)

Question 12

An edge of a discrete beam connects $\mathbf{x}_1 = (0, 0)$ and $\mathbf{x}_2 = (3, 4)$, with lengths in meters. Its undeformed length is $\Delta l = 4$ m and its stretching stiffness is $EA = 100$ N. What is the stretching energy $E_s = \frac{1}{2}EA\,\varepsilon^2\Delta l$, where $\varepsilon = \frac{|\mathbf{x}_2 - \mathbf{x}_1|}{\Delta l} - 1$?

(a) 50 J

(b) 3.125 J

(c) 25 J

(d) **12.5 J** ✓

Answer: (d)
Solution. $|\mathbf{x}_2 - \mathbf{x}_1| = \sqrt{3^2 + 4^2} = 5$ m, so $\varepsilon = 5/4 - 1 = 0.25$. $E_s = \frac{1}{2} \cdot 100 \cdot 0.25^2 \cdot 4 = \frac{1}{2} \cdot 100 \cdot 0.0625 \cdot 4 = 12.5$ J. (Option (b) forgets the factor Δl; option (c) forgets the $\frac{1}{2}$.)

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Question 13

An edge of a discrete rod goes from $\mathbf{x}_1 = (0,0,0)$ to $\mathbf{x}_2 = (0,0,2)$. Which of the following $(\mathbf{m}_1, \mathbf{m}_2, \mathbf{t})$ is an orthonormal adapted frame on this edge? (Recall that it must satisfy $\mathbf{m}_1 \times \mathbf{m}_2 = \mathbf{t}$.)

(a) $\mathbf{m}_1 = (1,0,0)$, $\mathbf{m}_2 = (0,-1,0)$, $\mathbf{t} = (0,0,1)$

(b) **$\mathbf{m}_1 = (0,1,0)$, $\mathbf{m}_2 = (-1,0,0)$, $\mathbf{t} = (0,0,1)$** ✓

(c) $\mathbf{m}_1 = (1,1,0)$, $\mathbf{m}_2 = (-1,1,0)$, $\mathbf{t} = (0,0,1)$

(d) $\mathbf{m}_1 = (1,0,0)$, $\mathbf{m}_2 = (0,0,-1)$, $\mathbf{t} = (0,1,0)$

Answer: (b)
Solution. The tangent is $\mathbf{t} = (0,0,2)/2 = (0,0,1)$. In (b), the three vectors are unit vectors, they are mutually orthogonal, and $\mathbf{m}_1 \times \mathbf{m}_2 = (0,1,0) \times (-1,0,0) = (0,0,1) = \mathbf{t}$. (a) gives $\mathbf{m}_1 \times \mathbf{m}_2 = (0,0,-1) = -\mathbf{t}$; (c) has vectors that are not unit length; (d) is a right-handed orthonormal frame ($\mathbf{m}_1 \times \mathbf{m}_2 = (0,1,0) = \mathbf{t}$), but it is not adapted, because its $\mathbf{t}$ is not along the edge.

Question 14

Why is a material frame (or, equivalently, a twist angle on each edge) needed in addition to the nodal coordinates to describe a 3D discrete elastic rod?

(a) **The nodal coordinates describe only the centerline; they cannot capture the rotation (twist) of the cross-section about the centerline.** ✓

(b) The nodal coordinates cannot describe the bending curvature of a rod in 3D.

(c) The material frame is needed to compute the stretching energy of each edge.

(d) Without a material frame, the Jacobian of the 3D rod is always singular, even when some DOFs are fixed.

Answer: (a)
Solution. The nodes describe only the centerline. Rotating the cross-section about the centerline (twisting the rod) leaves every node where it is but changes the elastic energy, so an extra quantity (the material frame, or the twist angle θ^k on each edge) is needed.

Question 15

A discrete rod has three nodes,

$$\mathbf{x}_0 = [0,\ 0,\ 0], \qquad \mathbf{x}_1 = [1,\ 0,\ 0], \qquad \mathbf{x}_2 = [1,\ 1,\ 0],$$

and two edges, $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{e}^1 = \mathbf{x}_2 - \mathbf{x}_1$. The vector $\mathbf{u} = [1,\ 0,\ 0]$ lies on the first edge. What is the result of parallel transporting $\mathbf{u}$ from the first edge ($\mathbf{e}^0$) to the second edge ($\mathbf{e}^1$)?

(a) $[1,\ 0,\ 0]$

(b) $[0,\ 0,\ 1]$

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(c) **[0, 1, 0]** ✓

(d) $[-1,\ 0,\ 0]$

Answer: (c)
Solution. $\mathbf{t}^0 = [1,0,0]$ and $\mathbf{t}^1 = [0,1,0]$. Following the steps in the notes: $\mathbf{b} = \mathbf{t}^0 \times \mathbf{t}^1 = [0,0,1] = \hat{\mathbf{b}}$, $\mathbf{n}_1 = \mathbf{t}^0 \times \hat{\mathbf{b}} = [0,-1,0]$, $\mathbf{n}_2 = \mathbf{t}^1 \times \hat{\mathbf{b}} = [1,0,0]$. With $\mathbf{u} = [1,0,0]$: $\mathbf{u}\cdot\mathbf{t}^0 = 1$, $\mathbf{u}\cdot\mathbf{n}_1 = 0$, $\mathbf{u}\cdot\hat{\mathbf{b}} = 0$, so $\mathbf{d} = 1\cdot\mathbf{t}^1 + 0 + 0 = [0,1,0]$. Here $\mathbf{u}$ is the tangent of the first edge, and parallel transport maps it to the tangent of the second edge.

Question 16

Which statement about parallel transport of a vector $\mathbf{u}$ (with $\mathbf{u} \perp \mathbf{t}^{k-1}$) from edge $\mathbf{e}^{k-1}$ to edge $\mathbf{e}^k$ is **FALSE**?

(a) The result is perpendicular to $\mathbf{t}^k$ and has the same length as $\mathbf{u}$.

(b) If $\mathbf{t}^k = \mathbf{t}^{k-1}$, the result is $\mathbf{u}$ itself.

(c) It is equivalent to rotating $\mathbf{u}$ about the axis $\mathbf{t}^{k-1} \times \mathbf{t}^k$ by the angle between $\mathbf{t}^{k-1}$ and $\mathbf{t}^k$.

(d) **It rotates u about the tangent $\mathbf{t}^k$ by the integrated twist τ_k.** ✓

Answer: (d)
Solution. Parallel transport is the twist-free way of moving a vector from one edge to the next: it rotates about $\mathbf{t}^{k-1} \times \mathbf{t}^k$ (statement (c)) and adds no rotation about the tangent. So (d) is false. Statements (a)–(c) are true.

Question 17

Two consecutive edges of a rod are collinear, with $\mathbf{t}^1 = \mathbf{t}^2 = (1,0,0)$. The first material directors are $\mathbf{m}_1^1 = (0,1,0)$ and $\mathbf{m}_1^2 = (0,0,1)$. The integrated twist τ_2 is the signed angle from $P_1^2(\mathbf{m}_1^1)$ to $\mathbf{m}_1^2$ about $\mathbf{t}^2$. What is τ_2?

(a) **$\pi/2$** ✓

(b) $-\pi/2$

(c) 0

(d) π

Answer: (a)
Solution. The tangents are equal, so parallel transport does nothing: $P_1^2(\mathbf{m}_1^1) = \mathbf{m}_1^1 = (0,1,0)$. The angle between $(0,1,0)$ and $(0,0,1)$ is $\pi/2$, and $(0,1,0) \times (0,0,1) = (1,0,0) = +\mathbf{t}^2$, so the sign is positive: $\tau_2 = \pi/2$.

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Question 18

In the DER algorithm (time-parallel reference frame), the twist angles on two consecutive edges are $\theta^1 = 0.1$ rad and $\theta^2 = 0.4$ rad, and the integrated reference twist at node $\mathbf{x}_2$ is $\Delta m_{2,\text{ref}} = -0.1$ rad. What is the integrated discrete twist τ_2?

(a) 0.4 rad

(b) **0.2 rad** ✓

(c) 0.3 rad

(d) 0.5 rad

Answer: (b)
Solution. $\tau_2 = (\theta^2 - \theta^1) + \Delta m_{2,\text{ref}} = (0.4 - 0.1) + (-0.1) = 0.2$ rad. (Option (c) ignores the reference twist.)

Question 19

At node $\mathbf{x}_2$ of a discrete rod, the integrated twist is $\tau_2 = 0.5$ rad and the twisting stiffness is $GJ = 2$ N·m^2. The undeformed lengths of the two neighboring edges are $|\bar{\mathbf{e}}^1| = 1$ m and $|\bar{\mathbf{e}}^2| = 3$ m. What is the twisting energy $E_2^t = \frac{1}{2} GJ \tau_2^2 \frac{1}{\bar{l}_2}$?

(a) 0.25 J

(b) 0.0625 J

(c) **0.125 J** ✓

(d) 0.5 J

Answer: (c)
Solution. The Voronoi length is $\bar{l}_2 = (1 + 3)/2 = 2$ m. $E_2^t = \frac{1}{2} \cdot 2 \cdot 0.5^2 \cdot \frac{1}{2} = 0.125$ J.

Question 20

A discrete rod has three nodes, $\mathbf{x}_0 = [0, 0, 0]$, $\mathbf{x}_1 = [1, 0, 0]$, and $\mathbf{x}_2 = [1, 1, 0]$ (in meters), and two edges, $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{e}^1 = \mathbf{x}_2 - \mathbf{x}_1$. The material frames on the two edges are

$$\mathbf{m}_1^0 = [0, 0, 1],\ \mathbf{m}_2^0 = [0, -1, 0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [1, 0, 0],\ \mathbf{m}_2^1 = [0, 0, -1] \ \text{ on } \mathbf{e}^1.$$

The integrated twist τ_1 at node $\mathbf{x}_1$ is the signed angle from $P_0^1(\mathbf{m}_1^0)$ (the parallel transport of $\mathbf{m}_1^0$ from $\mathbf{e}^0$ to $\mathbf{e}^1$) to $\mathbf{m}_1^1$, about the axis $\mathbf{t}^1$. What is τ_1?

(a) 0

(b) $-\pi/2$

(c) π

(d) $\boldsymbol{\pi/2}$ ✓

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Answer: (d)
Solution. $\mathbf{t}^0 = [1,0,0]$ and $\mathbf{t}^1 = [0,1,0]$, so $\hat{\mathbf{b}} = [0,0,1]$. Since $\mathbf{m}_1^0 = [0,0,1]$ is along $\hat{\mathbf{b}}$, parallel transport leaves it unchanged: $P_0^1(\mathbf{m}_1^0) = [0,0,1]$. The angle between $[0,0,1]$ and $\mathbf{m}_1^1 = [1,0,0]$ is $\pi/2$, and $[0,0,1] \times [1,0,0] = [0,1,0] = +\mathbf{t}^1$, so $\tau_1 = +\pi/2$.

Question 21

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [1,0,0]$, and $\mathbf{x}_2 = [1,1,0]$ (in meters), and two edges, $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{e}^1 = \mathbf{x}_2 - \mathbf{x}_1$. The material frames on the two edges are

$$\mathbf{m}_1^0 = [0,0,1],\ \mathbf{m}_2^0 = [0,-1,0] \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [1,0,0],\ \mathbf{m}_2^1 = [0,0,-1] \text{ on } \mathbf{e}^1.$$

The rod is naturally untwisted, both edges have undeformed length 1 m, and the twisting stiffness is $GJ = 1$ N·m^2. What is the twisting energy $E_1^t = \frac{1}{2} GJ \tau_1^2 \frac{1}{\bar{l}_1}$ at node $\mathbf{x}_1$?

(a) **$\boldsymbol{\pi^2/8}$ J** ✓

(b) $\pi^2/4$ J

(c) $\pi/4$ J

(d) $\pi^2/16$ J

Answer: (a)
Solution. From the previous question, $\tau_1 = \pi/2$. The Voronoi length is $\bar{l}_1 = (1+1)/2 = 1$ m. $E_1^t = \frac{1}{2} \cdot 1 \cdot (\pi/2)^2 \cdot \frac{1}{1} = \pi^2/8$ J.

Question 22

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [1,0,0]$, and $\mathbf{x}_2 = [1,1,0]$ (in meters), and two edges, $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{e}^1 = \mathbf{x}_2 - \mathbf{x}_1$. The material frames on the two edges are

$$\mathbf{m}_1^0 = [0,0,1],\ \mathbf{m}_2^0 = [0,-1,0] \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [1,0,0],\ \mathbf{m}_2^1 = [0,0,-1] \text{ on } \mathbf{e}^1.$$

The (time-parallel) reference frames on the two edges are

$$\mathbf{a}_1^0 = [0,0,1],\ \mathbf{a}_2^0 = [0,-1,0] \text{ on } \mathbf{e}^0, \qquad \mathbf{a}_1^1 = \tfrac{1}{\sqrt{2}}[1,0,1],\ \mathbf{a}_2^1 = \tfrac{1}{\sqrt{2}}[1,0,-1] \text{ on } \mathbf{e}^1.$$

The reference twist $\Delta m_{1,\text{ref}}$ at node $\mathbf{x}_1$ is the signed angle from $P_0^1(\mathbf{a}_1^0)$ to $\mathbf{a}_1^1$, about the axis $\mathbf{t}^1$. What is $\Delta m_{1,\text{ref}}$?

(a) $-\pi/4$

(b) **$\boldsymbol{\pi/4}$** ✓

(c) $\pi/2$

(d) 0

Answer: (b)
Solution. $\mathbf{a}_1^0 = [0,0,1]$ is along $\hat{\mathbf{b}} = [0,0,1]$, so $P_0^1(\mathbf{a}_1^0) = [0,0,1]$. The angle to $\mathbf{a}_1^1 = \frac{1}{\sqrt{2}}[1,0,1]$: $\cos\alpha = [0,0,1] \cdot \frac{1}{\sqrt{2}}[1,0,1] = 1/\sqrt{2}$, so $\alpha = \pi/4$. Sign: $[0,0,1] \times [1,0,1] = [0,1,0]$, along $+\mathbf{t}^1$. So $\Delta m_{1,\text{ref}} = +\pi/4$.

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Question 23

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [1,0,0]$, and $\mathbf{x}_2 = [1,1,0]$ (in meters), and two edges, $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{e}^1 = \mathbf{x}_2 - \mathbf{x}_1$. The material frames on the two edges are

$$\mathbf{m}_1^0 = [0,0,1],\ \mathbf{m}_2^0 = [0,-1,0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [1,0,0],\ \mathbf{m}_2^1 = [0,0,-1] \ \text{ on } \mathbf{e}^1.$$

The (time-parallel) reference frames on the two edges are

$$\mathbf{a}_1^0 = [0,0,1],\ \mathbf{a}_2^0 = [0,-1,0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{a}_1^1 = \tfrac{1}{\sqrt{2}}[1,0,1],\ \mathbf{a}_2^1 = \tfrac{1}{\sqrt{2}}[1,0,-1] \ \text{ on } \mathbf{e}^1.$$

In the DER algorithm, the twist angle θ^k (a degree of freedom) on edge $\mathbf{e}^k$ is the signed angle from $\mathbf{a}_1^k$ to $\mathbf{m}_1^k$ about $\mathbf{t}^k$. What are θ^0 and θ^1?

(a) $\theta^0 = \pi/2$, $\theta^1 = \pi/4$

(b) $\theta^0 = 0$, $\theta^1 = -\pi/4$

(c) $\boldsymbol{\theta^0 = 0,\ \theta^1 = \pi/4}$ ✓

(d) $\theta^0 = 0$, $\theta^1 = \pi/2$

Answer: (c)
Solution. Edge 0: $\mathbf{a}_1^0 = \mathbf{m}_1^0 = [0,0,1]$, so $\theta^0 = 0$. Edge 1: $\cos\theta = \frac{1}{\sqrt{2}}[1,0,1] \cdot [1,0,0] = 1/\sqrt{2}$, so the angle is $\pi/4$. Sign: $[1,0,1] \times [1,0,0] = [0,1,0]$, along $+\mathbf{t}^1$. So $\theta^1 = +\pi/4$.

Question 24

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [1,0,0]$, and $\mathbf{x}_2 = [1,1,0]$ (in meters), and two edges, $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{e}^1 = \mathbf{x}_2 - \mathbf{x}_1$. The material frames on the two edges are

$$\mathbf{m}_1^0 = [0,0,1],\ \mathbf{m}_2^0 = [0,-1,0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [1,0,0],\ \mathbf{m}_2^1 = [0,0,-1] \ \text{ on } \mathbf{e}^1.$$

The (time-parallel) reference frames on the two edges are

$$\mathbf{a}_1^0 = [0,0,1],\ \mathbf{a}_2^0 = [0,-1,0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{a}_1^1 = \tfrac{1}{\sqrt{2}}[1,0,1],\ \mathbf{a}_2^1 = \tfrac{1}{\sqrt{2}}[1,0,-1] \ \text{ on } \mathbf{e}^1.$$

Compute the integrated twist at node $\mathbf{x}_1$ a second way, using the DER formula $\tau_1 = (\theta^1 - \theta^0) + \Delta m_{1,\text{ref}}$. What is τ_1?

(a) $\pi/4$

(b) $\boldsymbol{\pi/2}$ ✓

(c) 0

(d) $3\pi/4$

Answer: (b)
Solution. $\tau_1 = (\pi/4 - 0) + \pi/4 = \pi/2$. This matches the answer from parallel transporting the material frame (Question 20), as it must.

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Question 25

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [2,0,0]$, and $\mathbf{x}_2 = [2,3,0]$ (in meters). Both edges have undeformed length 2 m, and the rod is naturally straight ($\boldsymbol{\kappa}^0 = \mathbf{0}$). The material frames are

$$\mathbf{m}_1^0 = [0,1,0],\ \mathbf{m}_2^0 = [0,0,1] \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [-1,0,0],\ \mathbf{m}_2^1 = [0,0,1] \text{ on } \mathbf{e}^1.$$

The stiffnesses are $EA = 4$ N, $EI = 1$ N·m^2, and $GJ = 1$ N·m^2. What is the curvature binormal $(\kappa\mathbf{b})_1 = \dfrac{2\,\mathbf{t}^0 \times \mathbf{t}^1}{1 + \mathbf{t}^0 \cdot \mathbf{t}^1}$ at node $\mathbf{x}_1$?

(a) **[0, 0, 2]** ✓

(b) $[0, 0, -2]$

(c) $[0, 0, 1]$

(d) $[2, 0, 0]$

Answer: (a)
Solution. The unit tangents are $\mathbf{t}^0 = [1,0,0]$ and $\mathbf{t}^1 = [0,1,0]$ (the edge lengths do not matter). $\mathbf{t}^0 \times \mathbf{t}^1 = [0,0,1]$ and $\mathbf{t}^0 \cdot \mathbf{t}^1 = 0$, so $(\kappa\mathbf{b})_1 = 2[0,0,1]/(1+0) = [0,0,2]$.

Question 26

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [2,0,0]$, and $\mathbf{x}_2 = [2,3,0]$ (in meters). Both edges have undeformed length 2 m, and the rod is naturally straight ($\boldsymbol{\kappa}^0 = \mathbf{0}$). The material frames are

$$\mathbf{m}_1^0 = [0,1,0],\ \mathbf{m}_2^0 = [0,0,1] \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [-1,0,0],\ \mathbf{m}_2^1 = [0,0,1] \text{ on } \mathbf{e}^1.$$

The stiffnesses are $EA = 4$ N, $EI = 1$ N·m^2, and $GJ = 1$ N·m^2. The material curvatures at node $\mathbf{x}_1$ are $\kappa_1 = \frac{1}{2}(\kappa\mathbf{b})_1 \cdot (\mathbf{m}_2^0 + \mathbf{m}_2^1)$ and $\kappa_2 = -\frac{1}{2}(\kappa\mathbf{b})_1 \cdot (\mathbf{m}_1^0 + \mathbf{m}_1^1)$. What is $\boldsymbol{\kappa}_1 = [\kappa_1,\ \kappa_2]$?

(a) $[0,\ 2]$

(b) $[0,\ -2]$

(c) **[2, 0]** ✓

(d) $[1,\ 0]$

Answer: (c)
Solution. $\mathbf{m}_2^0 + \mathbf{m}_2^1 = [0,0,2]$, so $\kappa_1 = \frac{1}{2}[0,0,2] \cdot [0,0,2] = 2$. $\mathbf{m}_1^0 + \mathbf{m}_1^1 = [-1,1,0]$, so $\kappa_2 = -\frac{1}{2}[0,0,2] \cdot [-1,1,0] = 0$. So $\boldsymbol{\kappa}_1 = [2,0]$.

Question 27

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [2,0,0]$, and $\mathbf{x}_2 = [2,3,0]$ (in meters). Both edges have undeformed length 2 m, and the rod is naturally straight ($\boldsymbol{\kappa}^0 = \mathbf{0}$). The material frames are

$$\mathbf{m}_1^0 = [0,1,0],\ \mathbf{m}_2^0 = [0,0,1] \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [-1,0,0],\ \mathbf{m}_2^1 = [0,0,1] \text{ on } \mathbf{e}^1.$$

The stiffnesses are $EA = 4$ N, $EI = 1$ N·m^2, and $GJ = 1$ N·m^2. What is the bending energy $E_1^b = \frac{1}{2} EI\, |\boldsymbol{\kappa}_1 - \boldsymbol{\kappa}_1^0|^2\, \frac{1}{l_1}$ at node $\mathbf{x}_1$?

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(a) 2 J

(b) 0.5 J

(c) 4 J

(d) **1 J** ✓

Answer: (d)
Solution. The Voronoi length uses the undeformed edge lengths: $\bar{l}_1 = (2+2)/2 = 2$ m. With $\boldsymbol{\kappa}^0 = \mathbf{0}$: $E_1^b = \frac{1}{2} \cdot 1 \cdot (2^2 + 0^2) \cdot \frac{1}{2} = 1$ J.

Question 28

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [2,0,0]$, and $\mathbf{x}_2 = [2,3,0]$ (in meters). Both edges have undeformed length 2 m, and the rod is naturally straight ($\boldsymbol{\kappa}^0 = \mathbf{0}$). The material frames are

$$\mathbf{m}_1^0 = [0,1,0],\ \mathbf{m}_2^0 = [0,0,1] \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [-1,0,0],\ \mathbf{m}_2^1 = [0,0,1] \text{ on } \mathbf{e}^1.$$

The stiffnesses are $EA = 4$ N, $EI = 1$ N·m^2, and $GJ = 1$ N·m^2. What is the total stretching energy of the rod, $E^s = \sum_k \frac{1}{2} EA \left(\frac{|\mathbf{e}^k|}{|\bar{\mathbf{e}}^k|} - 1 \right)^2 |\bar{\mathbf{e}}^k|$?

(a) 0.5 J

(b) 2 J

(c) 4 J

(d) **1 J** ✓

Answer: (d)
Solution. Edge 0: $|\mathbf{e}^0| = 2 = |\bar{\mathbf{e}}^0|$, so its energy is 0. Edge 1: $|\mathbf{e}^1| = 3$, so $|\mathbf{e}^1|/|\bar{\mathbf{e}}^1| - 1 = 0.5$ and its energy is $\frac{1}{2} \cdot 4 \cdot 0.5^2 \cdot 2 = 1$ J. Total: $E^s = 1$ J.

Question 29

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [2,0,0]$, and $\mathbf{x}_2 = [2,3,0]$ (in meters). Both edges have undeformed length 2 m, and the rod is naturally straight ($\boldsymbol{\kappa}^0 = \mathbf{0}$). The material frames are

$$\mathbf{m}_1^0 = [0,1,0],\ \mathbf{m}_2^0 = [0,0,1] \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [-1,0,0],\ \mathbf{m}_2^1 = [0,0,1] \text{ on } \mathbf{e}^1.$$

The stiffnesses are $EA = 4$ N, $EI = 1$ N·m^2, and $GJ = 1$ N·m^2. The rod is naturally untwisted. What is the total elastic energy (stretching + bending + twisting) of the rod?

(a) 1 J

(b) **2 J** ✓

(c) 3 J

(d) 0 J

Answer: (b)
Solution. Stretching: $E^s = 1$ J. Bending: $E^b = 1$ J (previous questions). Twisting: parallel transporting $\mathbf{m}_1^0 = [0,1,0]$ from $\mathbf{t}^0 = [1,0,0]$ to $\mathbf{t}^1 = [0,1,0]$ gives $[-1,0,0] = \mathbf{m}_1^1$, so $\tau_1 = 0$ and $E^t = 0$. Total: $1 + 1 + 0 = 2$ J.

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